



Where's fair value for EUR/CNY?

- The EU is currently in the middle of what has been described as '[China Shock 2.0](#)', with its trade deficit with China widening materially in recent years. There are multiple factors simultaneously at play, including relative energy prices and subsidy policies. Another factor is the exchange rate.
- At the recent G7 Summit, German Chancellor Friedrich Merz said that the European Union was competing with countries whose currencies were highly undervalued. While not explicitly mentioned, the spotlight has fallen most heavily on the CNY.
- In this special report we investigate yuan valuations against the euro. To ensure robustness, we employ ten different models to assess valuations in EUR/CNY. Extending the analysis, we also look to where our suite of models would put the hypothetical fair value for DEM/CNY. This helps to understand whether Germany faces a more acute competitiveness problem compared to the wider Euro area.
- Our central results indicate that at its peak last year, EUR/CNY overvaluation was likely around 20%. With its recent appreciation, the yuan remains around 15% undervalued against the euro on our central estimates. This level of undervaluation is similar to the extremes seen during 2005-08. However, during that period euro over-valuation was much more broad-based. All our models suggest that yuan undervaluation is more pronounced against a hypothetical Deutschmark, illustrative of how Germany's competitiveness challenge is larger than the wider Euro Area.

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Figure 1: CNY remains firmly undervalued against the euro, and screens even cheaper against a hypothetical Deutschmark

Type of valuation	Model	CNY valuation vs EUR	CNY valuation vs DEM
Absolute PPP	Big Mac Index	-46%	-52%
	OECD PPP	-34%	-38%
Absolute PPP - detrended for Balassa-Samuelson effect	Big Mac Index - detrended	-20%	/
	OECD PPP - detrended*	-18%	-22%
Relative PPP	Bilateral, PPI-based PPP*	-14%	-14%
	Bilateral, CPI-based PPP	-11%	-12%
Relative PPP adjusted for growth, terms of trade	Dbeer	-10%	/
External balance	FEER - full current account	-9%	-16%
	FEER - Goods and services (Setser-adjusted)	-9%	-14%
	FEER - Goods trade only, exc. 22 energy shock*	-10%	-13%
Average of preferred (*) models		-14%	

Source : Deutsche Bank, The Economist, OECD, Haver Analytics, Bloomberg Finance LP
Note: The EUR/CNH spot rate was taken as 7.75 for all the PPP results.



Introduction

The European Union is currently in the middle of what our economists have described as '[China Shock 2.0](#)', with its trade deficit with China widening materially in recent years.

There are many factors at play. Firstly, China has moved up the manufacturing value chain. China is now in direct trade competition with the EU in more sectors, with cars perhaps the most visible example. Secondly, price differentials have moved in China's favour and against Europe, which is itself a result of multiple factors. These include the larger shock to European energy costs after Russia's invasion of Ukraine, subsidies which are lowering China's production costs, and the exchange rate. The latter is the broad focus of this report, and has become an increasingly prevalent topic of discussion in recent weeks.

Indeed, at the G7 Summit a few weeks ago, German Chancellor Friedrich Merz [reportedly](#) said that the EU was competing against countries whose currencies were highly undervalued, thus further hurting the European Union's competitiveness problem.

In this report, we look at ten different long-term metrics to assess the undervaluation of the yuan against the euro and a hypothetical Deutschmark. We look at two different types of long-term currency valuation: Purchasing Power Parity (PPP) and external balance models. They range from the very simple, but accessible, Big Mac Index to a Fundamental Equilibrium Exchange Rate (FEER) model adjusting for idiosyncracies of the respective China and Euro Area trade data.

Our central results indicate that, despite its recent appreciation, the yuan remains around 15% undervalued against the euro. This compares to an undervaluation of over 20% twelve months ago. All our models suggest that the undervaluation of the yuan is more pronounced against a hypothetical Deutschmark than that of the euro, illustrative of how Germany's competitiveness challenge is larger than the wider Euro Area.

Figure 2: Euro Area trade deficit with China in goods is widening



Source : Deutsche Bank, Haver Analytics



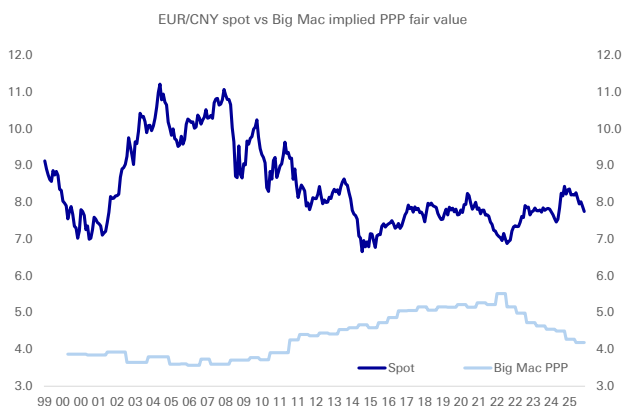
Purchasing Power Parity (PPP)

PPP models remain the bedrock of long-term currency valuation assessment, and use differences in prices to assess where nominal (spot) exchange rates should trade ("fair value").

We start with **absolute PPP**, and specifically with perhaps the simplest and most widely known model: [The Economist's Big Mac Index](#). Under the model, since a Big Mac costs around 25.5 yuan in China and €6.10 in the Euro Area (€6.90 in Germany), the *law of one price* in turn implies that fair value for EUR/CNY is just under 4.20 ([Figure 3](#)). With spot trading at around 7.75 at the time of writing, this very simple model suggests that CNY is around 45% undervalued against the euro.

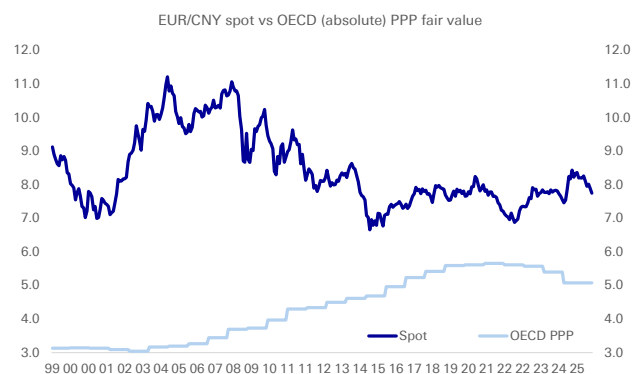
The OECD's PPP metric uses similar logic as the Big Mac Index, but looks at the relative price levels across a wide range of goods and services, rather than focusing on the relative cost of a single good.¹ Its results are similarly stark, pointing to an approximate 35% undervaluation of the yuan against the euro ([Figure 4](#)).

Figure 3: The relative price of a burger suggests the yuan is close to 50% undervalued against the euro



Source : Deutsche Bank, Haver Analytics

Figure 4: OECD measures - aggregated up at economy-wide scale - also imply material undervaluation



Source : Deutsche Bank, Haver Analytics

However, plotting the misalignments from these metrics over time is informative in several ways ([Figure 5](#) and [Figure 6](#)). Firstly, the yuan has *always* screened cheap on these metrics, showing between 20% and 70% undervaluation. Nevertheless, the undervaluations on these metrics had – until relatively recently – been firmly trending in a positive direction. In other words, CNY was becoming less undervalued over time. This is consistent with Balassa-Samuelson effect, a country like China with high productivity growth in its tradable goods sector should see its real exchange rate strengthen over time - leaving it less undervalued on absolute PPP metrics. However, this trend notably reversed over the past four years, which have seen a clear break in CNY valuation from this trend with valuations cheapening again since 2022.

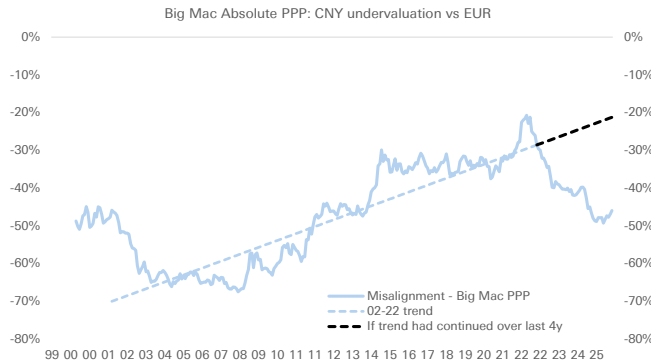
To adjust for these factors, [Figure 7](#) shows *detrended* Big Mac and OECD PPP valuations. These measures help illustrate the sharp cheapening over recent years,

1 Note: The OECD metric adjusts for each country's specific consumption and production basket, looking at inputs across GDP specific to that country, rather than a uniform basket



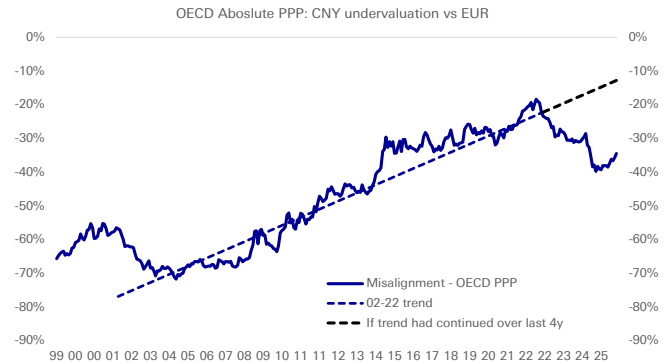
which even after this year's partial correction and fall in EUR/CNY spot, still sees the yuan around 20% undervalued against the euro, and cheaper still against a hypothetical Deutschmark, using German price data ([Figure 8](#)).

Figure 5: CNY had trended less cheap over time on Big Mac Index...



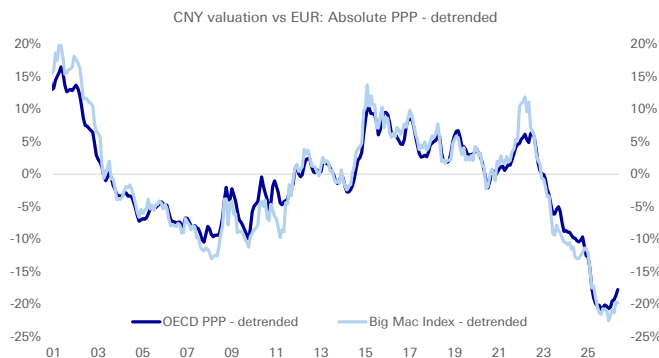
Source : Deutsche Bank, Haver Analytics

Figure 6: ... and on OECD PPP basis



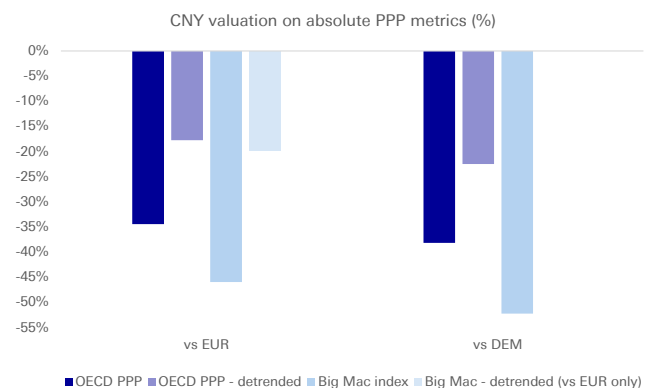
Source : Deutsche Bank, Haver Analytics

Figure 7: Detrended absolute PPP measures show around 20% cheapness in CNY vs EUR



Source : Deutsche Bank, Haver Analytics

Figure 8: Absolute PPP metrics suggest yuan even more undervalued against a hypothetical Deutschmark than euro



Source : Deutsche Bank, Haver Analytics

Absolute PPP imposes a strict condition for valuations by assuming that nominal exchange rates should move to offset specific *price level* differences. A softer variant is **relative PPP**, which instead assumes that exchange rates should move to offset differences in *changes* in prices (i.e., inflation) over time. In other words, the assumption is that real exchange rates are stationary.

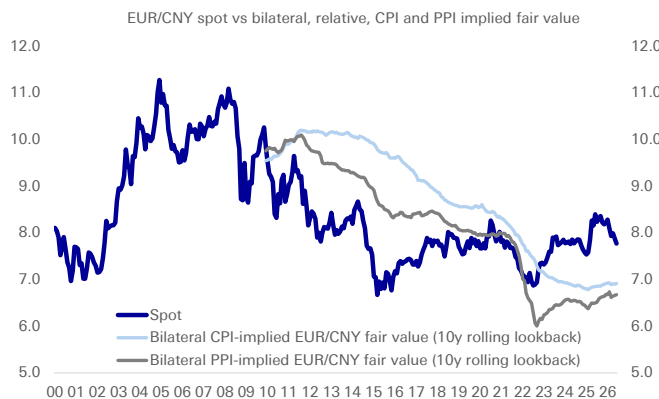
We build two simple relative PPP models. Firstly, we calculate the bilateral real exchange rate for EUR/CNY by deflating spot by the ratio of the respective consumer price indices (CPI). To then obtain our valuation, we simply compare this real exchange rate to its (rolling) 10-year average. We then repeat this methodology using producer prices (PPI), and also then again specifically using German rather than Euro Area-wide price data. The results over time are shown in [Figure 9](#) and [Figure 10](#). The PPI-metric currently points to a c. 14% undervaluation in the yuan against the euro, with similar results when looking just at Germany.



One thing to note is that this PPI-based metric implicitly assumes that the impact of any subsidies in China is *directly reflected* in lower producer prices. If, for example, manufacturers instead paid the market rate for their inputs but were then later reimbursed via these subsidies, then the "true" undervaluation of the yuan on this metric would be larger in magnitude.

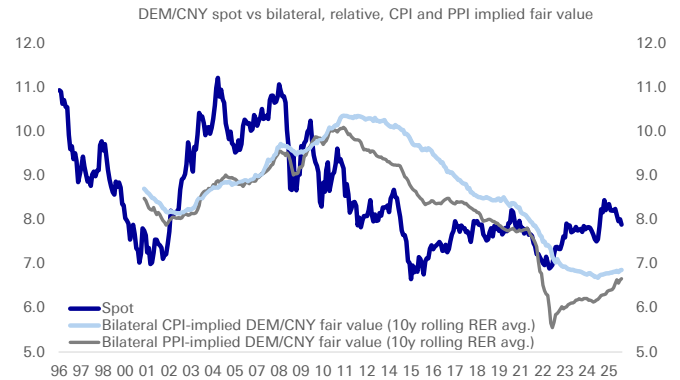
Going one step further, we can add some "extra trimmings" to our relative (CPI-based) PPP model, to control for other long-term drivers of real exchange rates such as productivity differentials, terms of trade changes, and economic openness. The implied fair value on this metric – our DBBeer model – is just over 7.15, suggesting CNY is around 8% undervalued against the euro ([Figure 11](#)).²

Figure 9: EUR/CNY spot vs CPI and PPI-implied relative PPP fair value



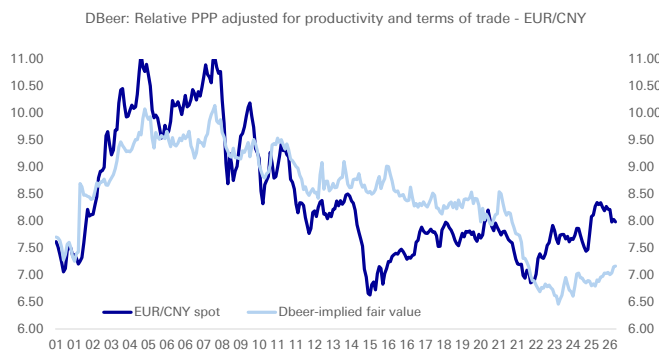
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 10: DEM/CNY spot vs CPI and PPI-implied relative PPP fair value



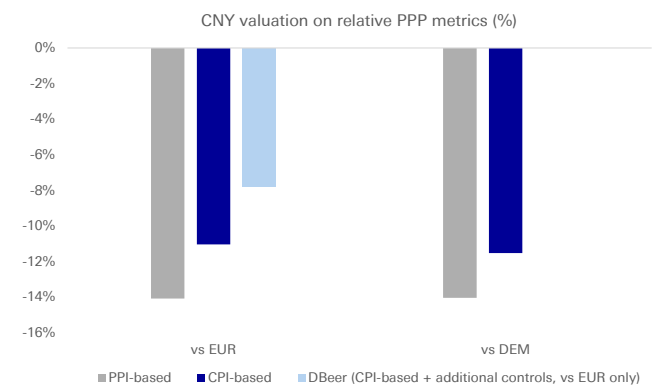
Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 11: EUR/CNY vs DBBeer (relative PPP adjusted for terms of trade and productivity)



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics

Figure 12: Overall yuan screens c. 12% cheap against the euro and c. 15% cheap vs DEM on relative PPP metrics



Source : Deutsche Bank, Bloomberg Finance LP, Haver Analytics. Note: EUR/CNH spot rate taken as 7.75 in these results

2 The full methodology for DBBeer can be found in [this paper](#)



External balance models

The PPP models described above all focus on the exchange rate movements required to equalise price levels or to offset inflation differences between regions. By contrast, external balance models estimate the exchange rate movement required to drive a single region's current account or trade balance to a specific level or "equilibrium".

Our benchmark external balance model, FEER, uses full current account data as its key input. This in turn comprises of trade in goods and services as well as income balances which capture cross-border investment returns (e.g. dividends) and transfer payments. Specifically, our benchmark model estimates the currency movement required to return a regions latest (last 12m) current account balance to its 10-year average, which we use as our proxy for the region's "equilibrium". To calculate the valuation, we use the average across two different estimates of current account sensitivity to exchange rates ("elasticity"), one from the IMF and one from Cline and Williamson (2008).

So, what do the latest results on our benchmark FEER model show? Overall, the full current account (inclusive of income balances) for the entire Euro Area is quite close to its average over the past decade (+1.7% of GDP vs 10y avg. +1.9%). To be sure, this average includes a period (2022-23) when the large energy shock put huge downward pressure on the goods balance, in particular. Nevertheless, as a result of the latest reading sitting only slightly below its average, on an aggregate trade-weighted (TWI) basis this baseline model only suggests a mild (<1%) overvaluation for the Euro. Meanwhile, China's headline current account balance sits at just under 4% of GDP, compared to an average over the last decade of c. 1.8% of GDP. As a result, our benchmark FEER model points to the renminbi being c. 7.5% undervalued on a broad, TWI, basis.

However, the headline current account data for both regions notably contain significant distortions. On the Euro Area side, the headline current account data are distorted by Ireland – see [here](#) for more. On the China side, [Setser](#) has noted the potential distortions to both the goods trade data flowing into the reported Balance of Payments figures and to the (investment) income side. To account for these, we also consider a few variations of our FEER model.

Firstly, we exclude income balances and focus just on the trade component of the current account, which captures both goods and services trade. We also use Setser's customs-based adjustment to the goods trade for China to account for the apparent change in methodology and reporting over the past few years. And on the Euro Area side, we exclude Irish trade from the data.

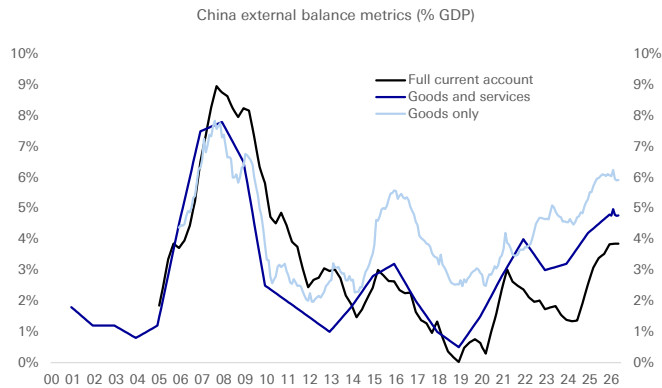
Secondly, we isolate trade in goods only. Here, for China, we take an average of three different reporting sources of Chinese goods trade data (domestic statistics body, WTO, IMF). All three broadly point to a surplus in goods trade alone of around 6% of GDP, vs a 10-year average of closer to 4% ([Figure 13](#)). For the Euro Area here, we again exclude Ireland, and we reduce the effect of the energy shock by assuming an unchanged goods balance between early 2022 and mid 2023. The gap between the Euro Area's latest external balance metrics and its 10-year average is most pronounced here, when looking only at goods trade (0.3% of GDP vs 0.9% average) ([Figure 14](#)).

Finally, we also look at external balance metrics for Germany specifically ([Figure 15](#)). Interestingly, the gap between Germany's latest external balance and its



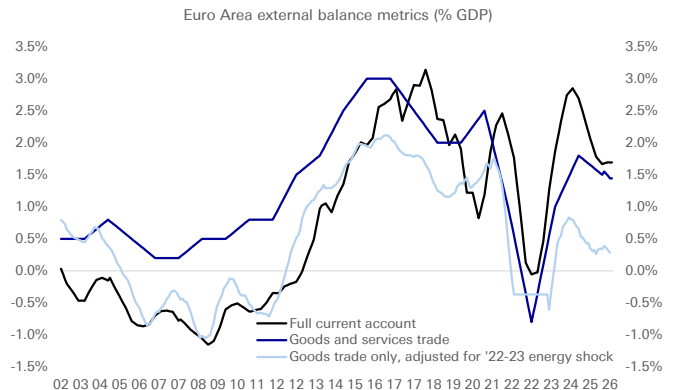
historical average widens when factoring in both services and income balances (full current account at 4.3% of GDP vs 6.8% average), compared to when solely looking at trade in goods (4.5% of GDP vs 6.0% average). [Figure 16](#) summarises the trade and external balance metrics used across our FEER iterations.

Figure 13: China external balances roughly 2ppt of GDP better than average, across multiple metrics



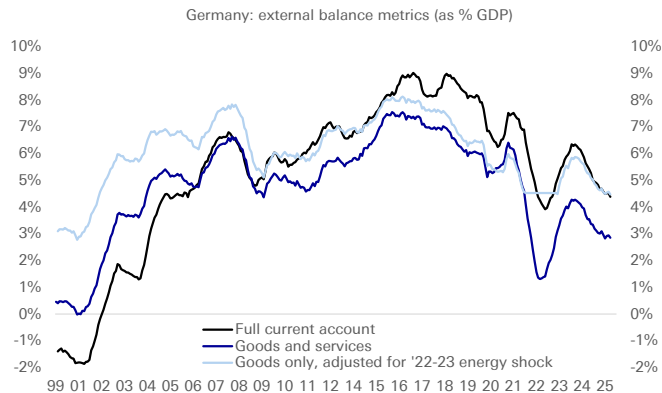
Source : Deutsche Bank, Haver Analytics, CFR

Figure 14: Euro Area wide external balances a little worse than average



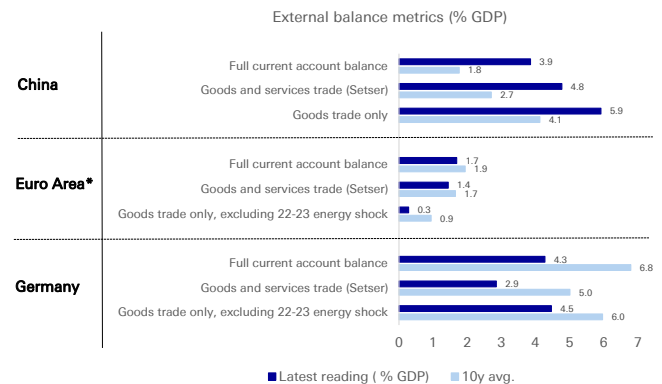
Source : Deutsche Bank, Haver Analytics

Figure 15: German external balance metrics firmly worse than historical average



Source : Deutsche Bank, Haver Analytics

Figure 16: Summary of external balance metrics for China, Euro Area, and Germany



Source : Deutsche Bank, Haver Analytics

*Euro Area exc. Ireland for goods and services and goods only iterations

[Figure 17](#) plots the results over time for China on our three different FEER iterations. All produce similar results, showing the yuan to be close to its historical extremes of undervaluation in trade-weighted terms. For Europe, all three iterations of the model indicate only a very mild overvaluation for the euro as a whole. However, the results are more striking when looking at Germany specifically. Using goods trade only, a hypothetical Deutschmark is just over 4% overvalued on a broad basis.³ However, its overvaluation doubles when looking at the whole German current account, inclusive of services trade and income balances ([Figure 18](#)).

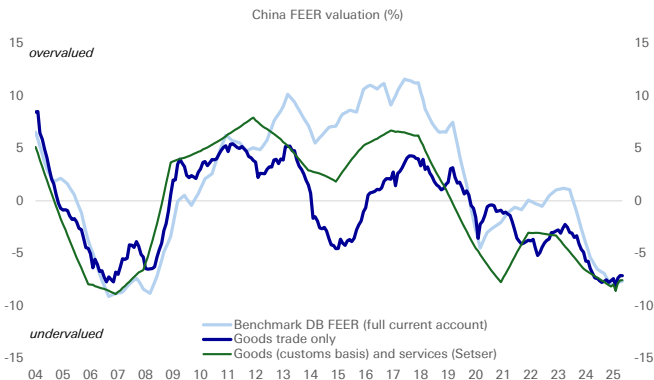
³ We apply the Euro Area's trade weights and elasticities here for Germany



FEER metrics are generally used for partial equilibrium analysis for one region's trade with the rest of the world, as opposed to bilateral analysis between two specific countries. As such, results are usually presented in terms of trade-weighted currency valuations (TWI, rather than specific crosses. Nevertheless, to allow for comparison with our other metrics, we use the matrix inversion method ([Cline](#)) to convert our implied broader TWI valuations into EUR/CNY (& DEM/CNY) misalignments. [Figure 19](#) shows the results across the three FEER variations.

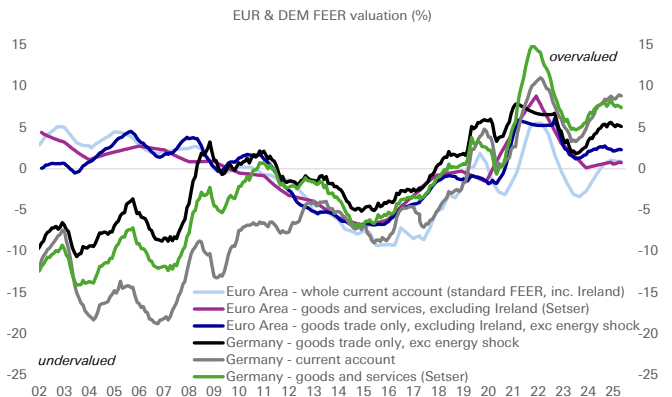
Unlike PPP metrics, the valuations are not directly affected by immediate changes in spot rates. Its valuations only change once the external balance metrics start to converge closer to their average and thus are slower-moving. As our [colleagues](#) note for Germany, there are some small signs of a nascent reversal in the worsening trade deficit with China ([Figure 20](#)). In part this may be because some of the strengthening over the past year in the yuan against the euro is starting to feed through. But as all our models here show, a large portion of the currency undervaluation remains.

Figure 17: Yuan undervaluation close to historical extremes across all iterations of our FEER models



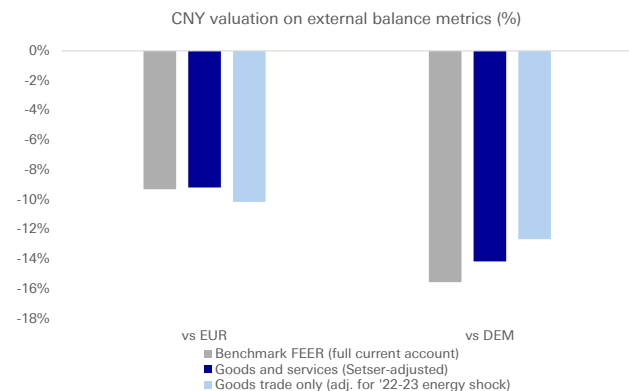
Source : Deutsche Bank, Haver Analytics

Figure 18: FEER models show more pronounced overvaluation for a hypothetical DEM than wider Euro Area



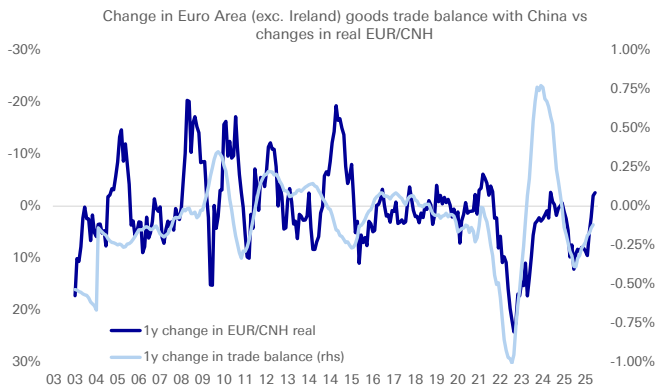
Source : Deutsche Bank, Haver Analytics

Figure 19: Implied CNY undervaluation vs EUR & DEM on FEER metrics



Source : Deutsche Bank, Haver Analytics

Figure 20: Nascent signs of slowing down in deterioration of Euro Area trade deficit with China



Source : Deutsche Bank, Haver Analytics



Conclusion

Our central results indicate that, despite its recent appreciation, the yuan remains around 15% undervalued against the euro. This compares to an undervaluation of over 20% twelve months ago. Nearly all our models suggest that the undervaluation of the yuan is more pronounced against a hypothetical Deutschmark than that of the euro, illustrative of how Germany's competitiveness challenge is larger than the wider Euro Area. To be sure, the many causes of Europe's competitiveness struggles vis-à-vis China are unlikely to all be solved by EUR/CNY moving closer to fair value. Nevertheless, they would be helped at the margin by the further strengthening in the yuan which our colleagues [expect](#).

Figure 21: Despite this year's rally, CNY remains undervalued against the euro, and screens even cheaper against a hypothetical Deutschmark

Type of valuation	Model	CNY valuation vs EUR	CNY valuation vs DEM
Absolute PPP	Big Mac Index	-46%	-52%
	OECD PPP	-34%	-38%
Absolute PPP - detrended for Balassa-Samuelson effect	Big Mac Index - detrended	-20%	/
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Average of preferred (*) models		-14%	

Source : Deutsche Bank, The Economist, OECD, Haver Analytics, Bloomberg Finance LP
Note: The EUR/CNH spot rate was taken as 7.75 for all the PPP results



Appendix 1

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*Other information available upon request

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